2 A student investigates light from different galaxies.

Fig. 2.1 shows the lines in the absorption spectrum from a distant galaxy.

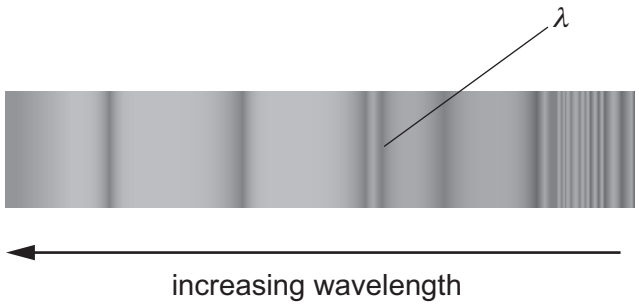


Fig. 2.1

The wavelength of one of the lines in the absorption spectrum is λ. The wavelength of this spectral line in the laboratory is λ_0.

The observations of the same spectral line are repeated for different galaxies.

The student determines the distance d of each galaxy from the Earth.

It is suggested that λ and d are related by the equation

$$\frac{\lambda - \lambda_0}{\lambda_0} = \frac{Hd}{c}$$

where c is the speed of light in free space and H is the Hubble constant.

(a) A graph is plotted of λ on the y-axis against $\frac{d}{c}$ on the x-axis.

Determine expressions for the gradient and y-intercept.

gradient = ...

y-intercept = ...

[1]

DO NOT WRITE IN THIS MARGIN

(b) Values of d and λ are given in Table 2.1.

Table 2.1

$d/10^{21}\,\text{km}$	$\frac{d}{c}/10^{15}\,\text{s}$	λ/nm
0.48 ± 0.12		658.4
1.04 ± 0.12		661.2
1.45 ± 0.12		664.2
1.80 ± 0.12		665.7
2.85 ± 0.12		672.4
3.75 ± 0.12		678.2

The value of c is $3.00 \times 10^5\,\text{km}\,\text{s}^{-1}$.

Calculate and record values of $\frac{d}{c}/10^{15}\,\text{s}$ in Table 2.1. Include the absolute uncertainties in $\frac{d}{c}$. [2]

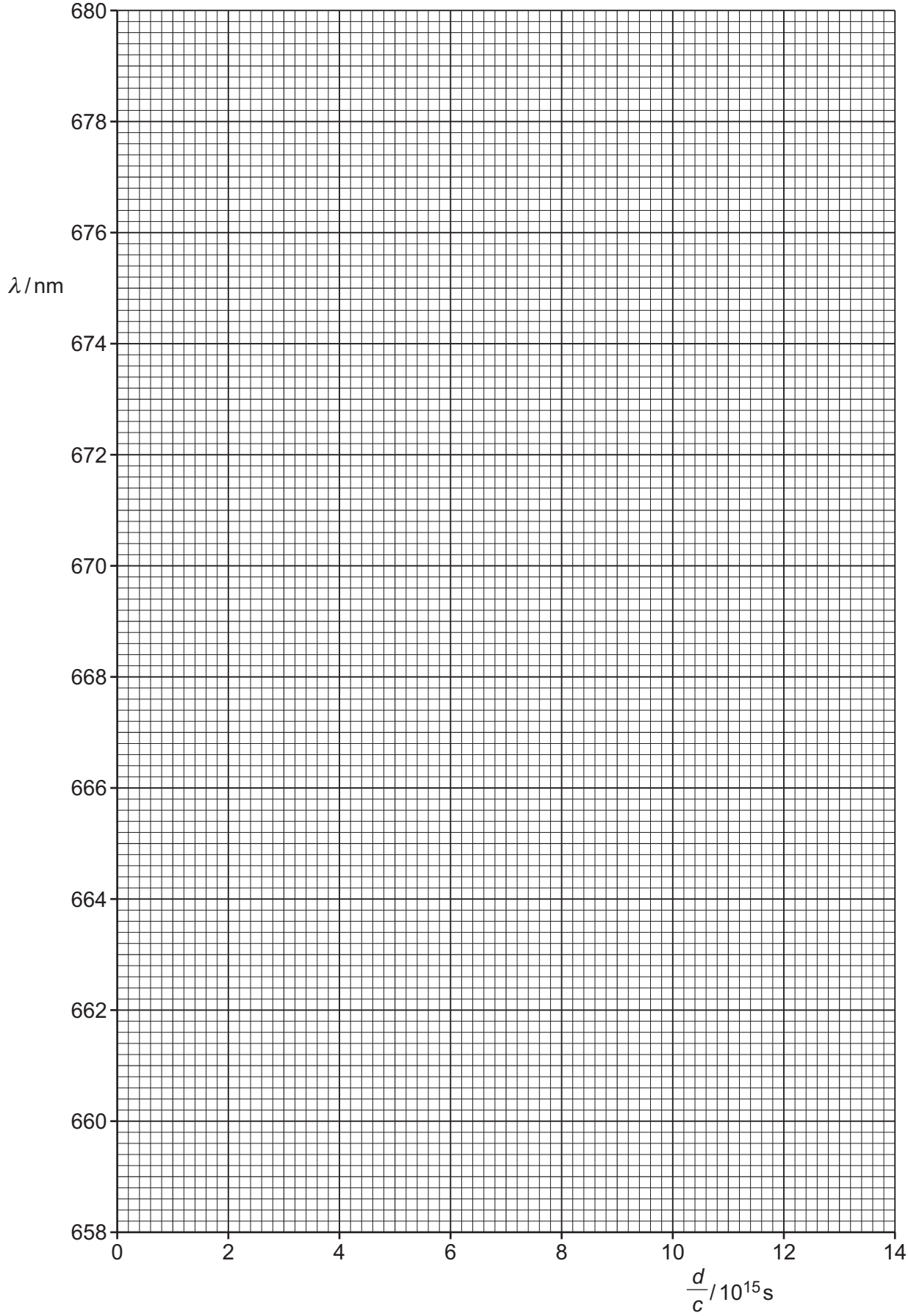
(c) (i) Plot a graph of λ/nm against $\frac{d}{c}/10^{15}\,\text{s}$. Include error bars for $\frac{d}{c}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]

DO NOT WRITE IN THIS MARGIN



DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(iv) Determine the *y*-intercept of the line of best fit. Include the absolute uncertainty in your answer.

y-intercept = ... [2]

(d) Using your answers to **(a)**, **(c)(iii)** and **(c)(iv)**, determine the values of λ_0 and H. Include appropriate units.

λ_0 = ...

H = ...

[2]

(e) Hubble's law suggests that the age T of the universe is related to H by

$$T = \frac{1}{H}.$$

Determine a value for T. Include the absolute uncertainty in your answer.

T = ... s [2]

[Total: 15]

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN